Importance of AI in healthcare

As per a report on Marketwatch.com, it is predicted that the applications of AI in the healthcare space will be worth 431.97 billion rupees by 2021, growing at a rate of about 40 per cent annually. Based on this growth of AI applications in healthcare, the doctor-patient ratio in India is expected to reach 6.9:1,000 by 2023, from its 2017 ratio of 4.8:1,000. The capability of AI applications to improve doctors' efficiency will help in tackling challenges like uneven doctor-patient ratio, by providing rural populations with high-quality healthcare, and training doctors and nurses to handle complex medical procedures.

When doctors are unable to diagnose your health issues, what can you do? Doctors are well-trained professionals but are still unable to diagnose properly at times. At such times, AI can help make better decision by triaging (evaluation and characterisation) of the sick patients efficiently, and thus save lives.

With AI applications, decision support systems sift and decipher a large amount of data available within electronic medical records. Also, along with imaging technology, AI helps in detecting uncertainty by highlighting problem areas, thus improving the diagnosis process.

AI improves the interaction between healthcare providers and patients. AI-powered smart agents can search, find, present and apply the most current clinical knowledge in partnership with physicians and researchers, thus significantly improving clinical efficiency and quality of care.

Physicians' workload includes repetitive, tedious tasks involved in researching, diagnoses and analysing patient data. Traditionally, to interpret an X-ray image, a radiologist looks at hundreds of images and manually finds the best match for their X-ray. Now, AI is automatically able to detect the best match along with some slightly differing ones, too.

Examples of AI deployment for healthcare

AI, drones, robots, telemedicine and Big Data are all being employed in Wuhan hospitals to tackle the ongoing novel coronavirus (COVID-19) outbreak in China. AI-powered robots are being deployed at the front lines to contain spread of the virus. They are being used for disinfection, street patrols, food and medicine delivery in quarantine wards.

Many hospitals around the world are now using various AI-based solutions. For instance, AI-based delivery robots are being used to move around the hallways delivering pills, bringing lunch to patients and ferrying specimen and medical equipment to different labs. Some hospitals employ robotic nurses to deal with any surge of patients and consequent shortage of human nurses.

Montefiore Medical Center in New York City runs its various machine learning (ML) models on a variety of Intel hardware and optimised software, enabling the flexibility needed to accommodate a variety of use cases. One of these models determines which patients are at high risk of requiring intubation (tube insertion), enabling clinicians to take the necessary actions.

GE Healthcare is working on deep-learning-based X-ray image analysis for detecting pneumothorax (collapsed lung). With its Optima XR240amx mobile X-ray systems, images are captured to notify the possibility of pneumothorax to radiologists within seconds.

In Japan, AI-based humanoid robots are providing companionship to old and sick people, and are helping with various chores at home. These robots are capable of delivering food, picking up objects from around the house, finding things that may have been misplaced, and more.

Applications of AI include wearable gloves that can detect epileptic seizure. AI-powered robots work as virtual assistants for doctors and, in some cases, display diagnostic abilities with similar degree of accuracy as medical professionals.

AI in Indian healthcare industry

The use of AI in the Indian healthcare industry is diverse across sub-sectors. The industry comprises a number of segments including hospitals, pharmaceuticals, diagnostics, medical equipment and supplies, medical insurance, and telemedicine. AI solutions can be employed in all these segments.

In a country like India, with a large population, there are very few doctors as compared to the patients. AI can help tackle such uneven doctor-to-patient ratio problems. It can even tackle the problems of untrained doctors and nurses for complex procedures, and absence of healthcare access for remote and rural areas.

The government of India is taking keen interest in finalising the national strategy on AI and other emerging technologies. Its think-tank, NITI Aayog, has identified five such sectors, including healthcare, for implementation of AI.

NITI Aayog, in partnership with Google, is developing AI-based solutions to accelerate the future of ML and AI in India. The partnership will build AI-assisted models to detect diabetic retinopathy screening models that will help in early risk detection.

NITI Aayog has also signed statement of intent with various companies, including ABB, IBM and SAP, related to AI projects. It is also collaborating with Intel and Tata Institute of Fundamental Research on various AI-based research projects.

AI in telemedicine

The advent of AI, ML and automation is making remote patient monitoring through telemedicine a reality. Telemedicine and AI both help in cutting costs and diagnosing illnesses faster and more accurately.

Telemedicine is useful for follow-up visits, management of chronic conditions, medication management, specialist consultation and other clinical services that can be provided remotely via secure video and audio connections. This saves time and costly infrastructure, and serves to provide isolated communities with speedy delivery of medical expertise.

Telemedicine is growing, and so is the demand for it. In many countries